

Sound Affects

A User's Guide

Edited by
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BLOOMSBURY ACADEMIC
NEW YORK • LONDON • OXFORD • NEW DELHI • SYDNEY

BLOOMSBURY ACADEMIC
Bloomsbury Publishing Inc
1385 Broadway, New York, NY 10018, USA
50 Bedford Square, London, WC1B 3DP, UK
29 Earlsfort Terrace, Dublin 2, Ireland

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First published in the United States of America 2023

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Library of Congress Cataloging-in-Publication Data

Names: Mee, Sharon Jane, editor. | Robinson, Luke, editor.

Title: Sound affects: a user's guide / edited by Sharon Jane Mee and
Luke Robinson.

Description: New York : Bloomsbury Academic 2023. | Series: Thinking media
| Includes bibliographical references and index. | Summary: "A philosophical
analysis of sonically charged concepts to map a theory of "sound affects."—
Provided by publisher.

Identifiers: LCCN 2022025106 (print) | LCCN 2022025107 (ebook) |
ISBN 9781501388880 (hardback) | ISBN 9781501388927 (paperback) |
ISBN 9781501388897 (epub) | ISBN 9781501388903 (pdf) | ISBN 9781501388910

Subjects: LCSH: Sound (Philosophy) | Sound in motion pictures.

Classification: LCC B105.S59 S68 2023 (print) | LCC B105.S59 (ebook) |
DDC 534/.301–dc23/eng/20220705

LC record available at <https://lcn.loc.gov/2022025106>

LC ebook record available at <https://lcn.loc.gov/2022025107>

ISBN: HB: 978-1-5013-8888-0
ePDF: 978-1-5013-8890-3
eBook: 978-1-5013-8889-7

Series: Thinking Media

Typeset by Deanta Global Publishing Services, Chennai, India
Printed and bound in Great Britain

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Feedback

Manuel 'Mandel' Cabrera Jr.

Acoustic feedback emerges within a zone of precarity: a situation in which the possibilities for sounding and listening, for act and affect, become fraught with Damoclean dangers. We're moving a microphone here and there near some speaker, curating the sonic space around us to suit the needs of the moment, when suddenly that threatening whine arises, and we know we're in the zone of precarity. The sonic situation emerges, not so much as an object of manipulation but rather as something to which we're vulnerable and thus must cater, lest the tiniest of movements take the subtlest of sounds and magnify them until they spiral into intolerable shrieks. Typically, then, we withdraw in a rush to escape the zone, trying to restore our sense of mastery. We swing the mic away from the speaker and hurriedly adjust settings as sound suddenly strikes us as a predator in hiding, waiting to leap out in an instant and stab at our ears.

The zone's material principle is the escalating cycle. Sound is received and transduced into electronic signals, then amplified and transduced back into sound, and run back through the chain again and again in a so-called reinjection cycle but one in which the signal intensifies with each of the many trips through the loop. The nature of this sort of cycle is highly variable. The mechanical and electromagnetic properties of the equipment, the placement of transducers and the resonant qualities of the surrounding environment all condition the zone's contours: which movements, sounds and settings will put us at risk of uncontrollable escalation, and which ones will take us closer to safety. It's uncertainty of these conditions that makes the zone precarious. They are nuanced, dynamic and difficult to discern, so that when we're in it we're often probing in ignorance.

For some, though, the zone has nevertheless had its seductions. What began as an irritating side effect of the high-powered microphones, pickups, amplifiers and loudspeakers that became commercially available in the mid-twentieth century has ended up being, for musicians of many different stripes, a seductive source of new sounds. By using those sounds, they've taken up manifold relations to the zone of precarity: taming the escalating

cycle, toying with it, channelling its furious power, inhabiting it as curious explorers.

However, musicians have also sometimes used feedback as a medium in which to embrace precarity as such, of which feedback's zone of precarity serves as a synecdoche and expression. The vulnerability, the risk, the non-mastery entailed by being inhabitants of the world – these make us creatures of affect. This isn't simply true in the narrow sense that we're subjects of special, receptive experiences (e.g. perceptions and emotions) that arise through the world's impacts on us. Rather, it's true in a deeper sense, articulated by the composer Eliane Radigue in her reflections on the significance, not just of music, but of sound as such: that we are eddies in the flow of the world's becoming.¹

Although feedback issues from machines we've engineered to expand the scope of our control over sound, it emerges as what precisely exceeds and flouts that control. The form in which it does so is as primal howls, whines and growls that put us in touch with, as Radigue puts it, the 'breath, pulsation, beating, murmur'² that constitute the enormous vibratory continuum of which the audible range represents only an infinitesimal sliver. This continuum isn't merely one aspect of the world, as Radigue describes it. Rather, it's the oscillation of becoming itself – the intricately structured dynamism that suffuses everything around us as well as ourselves: what she calls 'the immense vibrating symphony of the universe', the 'cacophony punctuating the deep ever-present rhythm of the breath, pulsations, beating'.³ In feedback, we hear silence with the volume turned way up: aspects of the vibratory continuum that are normally undetectable by our ears – sounds too quiet to be heard, frequencies implicit in mechanical parts – magnified to audibility by the escalating cycle. Feedback thus confronts us with nature's feral riches – the exquisite complexity of becoming that always underlies the seemingly controllable, perceptible surface of things. In doing so, feedback discloses what Radigue has elegantly called, in the title of her remarkable essay, 'the mysterious power of the infinitesimal': a phrase that describes the inexhaustible dynamism of the world itself, which in feedback is dramatically taken out of hiding, arresting our ears in all its wild vibrancy.

Various artistic uses of feedback can, then, be seen as figures for our differing ways of responding to this dynamism, or the ways in which the world defies us and exceeds our grasp. Like Robinson Crusoe on the Island of Despair, some musicians enter the zone of precarity in a spirit of conquest, using the fine-grained techniques afforded us by audio technology and acoustic science to gain more and more meticulous understanding and control over what had previously seemed untamable. Others, however, have embraced non-mastery, that is, embraced affect. Among a number of other

examples of this approach I will discuss below, Radigue's own use of feedback as an instrument provides us with some of the finest examples. That is, her remarkable feedback compositions are expressive of the sense that the world isn't something we confront in a struggle for control, but rather the perpetual flow of becoming that makes us what we are – makes us beings of affect through and through.

Most people probably first encounter musical feedback in rock and roll. We see footage of a guitarist soloing in notes that scream in ecstasy, playing as if the dance of their fingers over the fretboard and their waves of the neck are summoning occult forces, even expressing the musician's erotic potency (e.g. in the case of male guitarists for whom the instrument becomes positively phallic). We hear feedback used as musical punctuation: to create aleatory moments that add drive or weightlessness in between explosions of rhythm section and drums. At a concert, we witness it made to swell into an overwhelming wall of sound bombarding our whole bodies, not just our ears. Since the 1960s, when it entered the repertoire of superstars like the Beatles, the Who, Jimi Hendrix, Jefferson Airplane and the Velvet Underground, feedback has become a central part of the rock lexicon – a vehicle for its aspirations to incantatory power.⁴

However, during the 1960s, feedback was also taken up by artists with closer ties to conservatories, universities and the art world than to commercialized popular music. In the wake of developments like the rise of *musique concrète* and Karlheinz Stockhausen's bold experiments in using sound equipment, electronic and acousmatic music were in this period at the top of the musical avant-garde's agenda. As with rock, the increasing commercial availability of cheap, sophisticated electronics only intensified this tendency.⁵ Musicians and composers laboured to craft themselves into acousticians and electrical engineers. They reconfigured sound equipment, scabbled for spare parts and improvised idiosyncratic assemblages in search of new kinds of sound. Their efforts were often shaped by John Cage's turn away from music as meticulously crafted creation towards an appreciation of the endless treasures contained in sound as such – one made possible by the surrender of will and the embrace of accident.⁶ Under this influence, many musicians and composers undertook experiments in search of sonic serendipity. It's no wonder, then, that many of them also became intrigued by the phenomenon of feedback.

Robert Ashley's *The Wolfman* (1964), one of the earliest feedback works to emerge from the avant-garde milieu of the 1960s, attempted to drag the audience kicking and screaming into the zone of precarity and deliver them over to feedback's predation. A single performer, clad in dark glasses and

presented in the guise of 'a sinister nightclub vocalist, spotlight and all',⁷ sings short phrases softly into a microphone as a tape piece assembled from processed found sound and electronic effects plays on the speaker system. However, the mic is fed into an amplifier cranked up to a level at which any sound causes feedback at incredibly high volumes, which is then modulated by the subtle changes in the shape of the performer's mouth as they vocalize, creating the impression that the feedback is swirling around the audience like a pack of screeching demons. This was years before rock music began its arms race towards increasingly rattling decibel levels and decades before artists like Merzbow and My Bloody Valentine cultivated reputations for live performances filled with deafening barrages of feedback and other noise. And so, Ashley's piece seemed *sui generis*, virtually unprecedented and became notorious.⁸ He cast the vocalist in the role of the titular Wolfman, a murderous lycanthrope masquerading as the most innocuous of performers – a lounge singer. In doing so, Ashley had taken the most viscerally pleasurable of art forms, music, and daringly made it into an assault, a painful ordeal.

Other artists, though, worked to distance feedback from its image as nothing but a sonic predator. Ashley's piece had demonstrated a keen interest in transgressing and thus refiguring the space of performance and audition. And, the piece's presentation of its performer as its audience's aggressor suggested an implicit violence in the traditional division between audiences as those subjected to sound and artists as those doing the subjecting. *The Wolfman*, then, set the stage for works that would situate both audiences and performers as equally subject to the escalating cycle's vagaries – witnesses to, rather than manipulators of, its unfolding. By doing so, these works afforded new kinds of relations to the zone of precarity that the escalating cycle creates and thus to what is at issue in how it affects us.

Perhaps the most well-known example in this regard is Steve Reich's *Pendulum Music* (1968). The instructions for the piece dictate that several mics be suspended by cables from the ceiling, each above a speaker lying on its back to which that mic is connected through an amplifier turned up to the point of feedback. The piece's performers then pull back and release the mics so that they're swinging freely over the speakers, at which point the performers 'sit down to watch and listen like the rest of the audience'.⁹ The result is a series of pulses, rhythmic and melodic but constantly changing according to the entropic nature of the process – the gradual slowing of the microphones' swings, which terminate with them at rest above the speakers, resulting in a hum of static feedback. Reich described *Pendulum Music* as 'the ultimate process piece' ... 'audible sculpture. If it's done right, it's kind of funny'.¹⁰ And its humour precisely comes from its narrative arc. As if giggling while goading a caged animal, *Pendulum Music* toys with feedback, repeatedly

drawing its attacks but pulling just out of reach. Over time, though, it's as if this playfully reckless spirit is worn down by the predator's tirelessness and collapses into inert submission. And yet, when it does so, it enacts a moment of discovery. What, in evasion, had seemed a threat is, in yielding, unveiled as the serenity of a drone. In this way, the piece stages a gradual shift from the performers' initial manipulation of the zone of precarity, through the slow evaporation of this manipulation's traces by entropy, until a static vibration envelops performers and audience alike – all meditatively passive to the zone's workings.

In other sound pieces from this period, we find similar attempts to transfigure feedback, to see past its air of menace, and use it to connect us to the complex dynamics linking performer, equipment, audience and space. Among these, for example, are the works of Alvin Lucier.¹¹ As a co-founding member of the Sonic Arts Union, in the 1960s and 1970s, Lucier witnessed many innovative uses of feedback among his collaborators. This included Ashley's *The Wolfman*, which Lucier himself performed on 7 January 1967 at Brandeis University; as well as pieces like David Behrman's *Wave Train* (1966) and Gordon Mumma's *Hornpipe* (1967).¹² Such efforts were echoed in Lucier's own early work, much of which made unique use of reinjection cycles – that is, feedback, albeit not always of the sort that involves continuous signal escalation.

Lucier used reinjection cycles, for example, in his *Music for Solo Performer* (1965) in which, at John Cage's encouragement, he generated sound through biofeedback.¹³ In this work, the performer is attached to electrodes that pick up their alpha brain waves. These signals are fed into loudspeakers and other transducers, whose agitations in turn activate percussion instruments. The resulting sounds are then heard by the performer, triggering brain waves that are fed back into the loop, so that their sensory and nervous systems become analogues for the microphone in acoustic feedback. Since alpha waves are autonomic responses – ones, in fact, requiring them to be as calm and still as possible – the performer is thus situated, not as the godlike creator of sounds, but simply as another element in the system, a participant on a par with the equipment and environment. Just as *Pendulum Music* stages the erasure of the boundary between performer and audience, *Music for Solo Performer* stages the erasure of boundary between performer and instrument. The performer becomes, not so much a creator of sounds, as one among many conduits through which the process unfolds.

In other pieces from this period, Lucier used a similar receptivity to achieve a kind of sonic palpation of the environment. For example, to perform his piece *Shelter* (1967), contact mics are placed on the walls, doorframes, and doors of an enclosed space (usually, a performance hall), allowing us

to hear sounds outside it as filtered through the resonant properties of the architecture: for example, footsteps on the sidewalk, animal cries, musicians practising in far corners of the building – whatever lies outside the walls of the venue.

Through this process, we indeed experience the venue as a shelter, but not because its walls seal us hermetically in, fortifying us against a hostile world outside. Rather, it's because we're made to feel the world as precisely what bounds us within where we are, holding us, cradling us in place there. In such pieces, sound becomes a means, neither to observing things from a distance nor to pleasuring ourselves aesthetically through shaping it at will, but rather to arousing in our bodies and throughout our being the revelation of our co-presence with things and space.¹⁴

This latter theme converged with Lucier's use of reinjection cycles in his acousmatic piece *I am Sitting in a Room* (1969). To make its original version, he recorded a short text describing the process behind it, which begins:

I am sitting in a room different from the one you are in now. I am recording the sound of my speaking voice and I am going to play it back into the room again and again until the resonant frequencies of the room reinforce themselves so that any semblance of my speech, with perhaps the exception of rhythm, is destroyed.¹⁵

As the piece progresses, Lucier's words grow ever more indistinct as they're repeatedly filtered through the room's material properties, recorded and reinjected in that recorded form back through the filter again. Gradually, it's as if the room itself begins speaking in its own voice of ethereal, undulating hums. Following Lucier's own description, the piece could be seen as the artist's erasure, accomplished through the disintegration of the speech by which he expresses his conception thereof. However, he goes on to describe the effect of this process as 'the natural resonant frequencies of the room articulated by speech'.¹⁶ Destruction and articulation, then, are bound into one. But this is no willful paradox. Rather, Lucier's voice is treated here as simply part of a duet that unfolds through its individual voices – this speech, that vibration of walls – enchained in a dialogue of transduction. In this dialogue, each ceases to be what it was as it's transfigured through interchange with the other. But it isn't thereby destroyed, or subjugated by violence, but precisely enabled and amplified. A duet isn't a battle for domination, after all. As one voice in the duet, then, Lucier's own expression is made possible by the room's resonance just as much as it's made possible by him – the perfect encapsulation of a line from William Carlos Williams that Lucier has taken up as a kind of slogan for his work: 'No ideas but in things'.¹⁷

In later works, Lucier redeploys the theme of mutually transformative interchange with an environment to refigure the reinjection cycle in its escalating forms. For example, in *Bird and Person Dyning* (1975), his first major use of acoustic feedback as a primary element, a performer with binaural microphones attached to their ears wanders in the space immediately surrounding a tiny electronic birdcall – originally, a Christmas ornament he'd received as a gift.¹⁸ The chirps are picked up by the microphones and played on loudspeakers, creating not only feedback but complex, spatialized forms that result from its heterodyning with the birdcalls. Lucier himself recounts his discovery of this effect: 'I began hearing phantom images of the birdcall, which seemed to come from inside my head and at the same time to be located in various parts of the room.'¹⁹ Here, the very body of the performer, and not just their voice and their ears, is inserted into the zone of precarity. Their resulting dialogue with the environment takes the form of their silent slow wandering on the stage, exploring but also shaping a sonically articulated space in which the distinction between outside and inside is blurred. Neither manipulators of feedback, nor victims of its violence, they become instead inhabitants of the escalating cycle, revealed as simply the interpenetration and co-determination of performer, objects and environment.

In these pieces from Reich and Lucier, we find no hint of feedback as a threat – one to be teased, deployed as a weapon, or appropriated as a force all one's own. Instead, it's figured in a different way altogether, for which there's no better image than Radigue's aforementioned 'the mysterious power of the infinitesimal.'²⁰ Radigue's own journeys with feedback took place in relative isolation. During the mid-1950s and again in the mid-1960s, she'd worked for both Pierre Schaeffer and Pierre Henry, assisting with their forays into *musique concrète*.²¹ She envisioned an entirely different kind of music of her own, though: a music of developments so slow as to be imperceptible, like the growing of plants.²² However, neither Schaeffer, Henry, nor the coterie that surrounded them took her seriously as a musician. As she says: 'I was the little intern who no one paid attention to, especially in such a macho environment. . . . To demonstrate to you how much my presence was appreciated in that place: one day, a technician, who I won't name, said "It's great to have Eliane in the studio because she makes it smell good!"'²³ Thus, when Radigue took up feedback as her primary medium while working at Henry's Studio Apsome, and presented him with the result, *Jouet Electronique* (Electronic Toy) (1967), Henry dismissed it out of hand.

But she persisted anyway. After leaving Studio Apsome, Radigue found herself in the situation of being a single mother without access to a studio,

left with only a few pieces of equipment Henry had bequeathed her: three tape recorders, a mixing board, an amplifier, two loudspeakers and a microphone.²⁴ Through meticulous, painstaking work, mostly done during the hours when her children were in school, she continued her tutelage in feedback, producing a series of acousmatic pieces assembled using acoustic and tape feedback, for example, $\Sigma = a = b = a + b$ (1969), *Usral* (1969), *Omnht* (1970), *Vice-Versa, etc. . .* (1970), *Stress-Osaka* (1970) and *Opus 17* (1970).²⁵

In Radigue's feedback compositions, we find a sense for the cosmic flow of becoming, and of our embeddedness therein. Hums and chirps, rumbles and throbbing. We could easily mistake them for celestial energies as detected by satellites and radio telescopes: the inferno of stars, the electromagnetic songs of planets, or the thrum of the universe's microwave background. These are vibrations that aren't produced for consumption in a moment, that don't cater to the time of anxious bustle, but instead change at the pace of the tide. If we're distracted for a minute, we can suddenly realize the ambience has shifted without being able to pin down how or when: hints of a glacial history that stretches beyond, but includes, human affairs – the time of nature's unfolding over epic spans.

In this way, Radigue soars past Lucier's concern for our immediate environments into one for the world as a whole. Both cross into the zone of precarity and allow themselves to be swept up in the escalating cycle, precisely to get in touch with our embeddedness in things. In *Usral*, *Omnht* and *Stress-Osaka*, which were composed to serve as elements in larger sculpture or installation pieces, Radigue, like Lucier, demonstrated a keen sensitivity to the place of the sounds she produced using the escalating cycle in the situation of sounding and listening.²⁶ However, whereas Lucier's pieces are content to explore the spaces and materials of equipment, architecture, and bodies, Radigue's reach for the totality of what lies beyond all this: the 'immense vibrating symphony' that is the world itself.

To access the large, though, she looks to the small. With a view to the large, we find, she argues, 'the air's powerful breath, violent intimidating tornados, deep dark waves emerging in long pulsations from cracks in the earth, joined with shooting fire in a flaming crackling' – in short, all of the forces our subjection to which embodies our precarity; our vulnerable, dependent place in things.²⁷ Beneath them, though, there's the common thread: the undercurrent of vibration, transduction, wordless dialogue, what Radigue calls 'the ever-present rhythm of the breath, pulsations, beating' whose presence throughout us and in all our surroundings constitutes our intimacy with the cosmic.²⁸ We emerge from this undercurrent, our power a bit of its own – its power for interchange with itself: as she puts it, 'An organ adapts itself to transformation of a miniscule zone from the immense

vibrating spectrum decoded into sounds captured, refined, meaningful, even as we begin our own 'crackling, roaring, howling and growling, the noises of life'.²⁹ From this power there issues our ordering of sounds in language and music: 'The subtle alchemy of sounds becomes, oh wonder, understood.'³⁰ Yet, she immediately reminds us, 'breath, pulsations, and beating remain' as 'natural harmonics unfurl into space in their own language.'³¹

For Radigue, feedback is simply one form in which this language of 'breath, pulsations, and beating' speaks through us, yet therefore exceeds us: the murmuring of the infinitesimal foundations of being – enabling our expression, yes, but never, ever as a monologue: 'The occasional accident, a disrupted relation between recorder-transmitter-recorder-playback, and there our medium assumes some independence. How, then, does it behave? Breath, pulsation, beating, sustained sound, depending on the mood. So much richness in all this "feedback" and other chance or provoked "interference".'³² To make contact with this murmuring, Radigue learned on her simple equipment how to play feedback, to let it play her. Endless hours over months and years, holding a microphone, patiently shifting it here and there by the tiniest of increments – a dance of near stillness whose moves were the subtlest of motions, her ears and hands letting themselves be choreographed by the most fragile of sounds. She entered into the zone of precarity, not to conquer it, toy with it or even just to explore it but to dwell there in contented equanimity, serving as a vehicle for the infinitesimal's untamed power.

In doing so, Radigue refigures the zone of precarity, and in the process the source and venue of precarity as such: the all-encompassing, ever-becoming world. The world becomes, not a space of danger, where anything that shapes us beyond our ken or control thereby enslaves and demeans us. Rather, it becomes the author of our being and the source of our power, with which we commune as much through accident as through deliberate effort. In apprenticing ourselves to it, we feel ourselves, not held hostage to it, but held in place by it, and our fragility as the conduit through which our own power is conferred: our power to embark upon 'adventures, explorations of this infinite mystery of the transmutation of noise into sound, of sound into music and, as with all true questions, to receive in response only a few "hows," never a "why," thus leaving endless freedom to trace one's path, to find one's voice.'³³

Notes

- 1 Eliane Radigue, 'The Mysterious Power of the Infinitesimal', *Leonardo Music Journal* 19 (2009): 47–9.
- 2 Radigue, 'The Mysterious Power of the Infinitesimal', 47.

- 3 Radigue, 'The Mysterious Power of the Infinitesimal', 47–8.
- 4 For example, the Beatles's hit 'I Feel Fine' was released in 1964, before any of these other bands exploded into popularity. 'I Feel Fine' begins with a brief moment of guitar feedback that John Lennon once claimed was 'the first feedback ever recorded' that wasn't 'on some old blues record from the Twenties'. Cf. John Lennon and Yoko Ono, 'Interview with David Sheff', *Playboy*, January 1981, <http://www.beatlesinterviews.org/dbjypb.int4.html> (accessed 15 September 2021).
- 5 For a comprehensive history of electronic music, including of the impact of the commercial availability of sound equipment on its development, cf. Thom Holmes, *Electronic and Experimental Music: Technology, Music and Culture* (New York: Routledge, 2016), especially ch. 1, 'Early History: Predecessors and Pioneers (1874–1960)', 1–166. More specific but not restricted to the phenomenon of feedback, for a historical survey of the use of microphones and speakers in electronic music, cf. Cathy van Eck, *Between Air and Electricity: Microphones and Loudspeakers as Instruments* (New York: Bloomsbury Academic, 2018).
- 6 For a substantial selection of John Cage's own reflections on the abrogation of intention in musical practice, which he associates with the phenomenon of silence, cf. John Cage, *Silence: Lectures and Writings* (Middletown, CT: Wesleyan University Press, 1973).
- 7 Robert Ashley, 'The Wolfman for Amplified Voice and Tape', in *Source: Music of the Avant-Garde 1966–73*, ed. Larry Austin and Douglas Kahn (Berkeley: University of California Press, 2011), 143–5; cf. 145. The two tape pieces Ashley used for performances of *The Wolfman* were *The Wolfman Tape* (1964), and later *The 4th of July* (1962): 143.
- 8 In his highly influential history of experimental music, the composer and music theorist Michael Nyman calls *The Wolfman* 'possibly the best-known feedback piece', and its influence extended beyond avant-garde and into popular music. For example, seeing a performance of *The Wolfman* has been cited as one of Iggy Pop's main inspirations for the bold rock experimentation he undertook with his seminal band the Stooges. Cf. Michael Nyman, *Experimental Music: Cage and Beyond* (New York: Cambridge University Press, 1999), 100–1; and Benjamin Piekut, *Experimentalism Otherwise: The New York Avant-Garde and Its Limits* (Berkeley: University of California Press, 2011), 18 and 185–6.
- 9 Steve Reich, 'Pendulum Music', in *Writings on Music, 1965–2000*, ed. Paul Hillier (New York: Oxford University Press, 2002), 32.
- 10 Steve Reich, interview with Thom Holmes, *OHM: The Early Gurus of Electronic Music*, www.furious.com/perfect/ohm/reich.html (accessed 12 February 2021).
- 11 I focus on Lucier only for reasons of space, and as one of the most well-known avant-garde composers who embraced non-mastery and coincidence in the use of reinjection cycles, and of feedback in particular. One

could easily approach similar themes through the work of other early feedback innovators. For example, there are Max Neuhaus's attempts at using feedback as a medium for enacting an immediate, non-linguistic form of community. Cf. his *Public Supply I* (1966) and *Max Feed* (1966), both outgrowths of his use of feedback in performances of John Cage's *Fontana Mix* (1958). Or, there's David Tudor's use of signal feedback in pieces like *Untitled* (1972), *Toneburst* (1975) and *Pulsers* (1976), which Tudor once described as entries in 'a never-ending series of discovered works in which electronic components are found to be natural objects'. Cf. David Tudor, *Pulsers/Untitled* (1984) [audio recording], USA: Lovely Music Ltd, liner notes.

- 12 Robert Ashley, 'Performances', *robertashley.org*, <http://www.robertashley.org/performances/performance-history.htm> (accessed 9 February 2021).
- 13 Lucier has recounted the importance of Cage's urging in the genesis of the piece, for whose first performance Cage served as an assistant, in a number of interviews and writings. For example, see Alvin Lucier, 'Alvin Lucier on a Lifetime of Experiments' [video], interviewer unlisted, Red Bull Music Academy, YouTube (uploaded 31 May 2017), https://www.youtube.com/watch?v=v-Pnb_ZE7Hs (accessed 8 February 2021).
- 14 One could also mention, for example, Lucier's *Vespers* (1968), in which the performers are sealed in a pitch-black room and left to navigate it using the clicks of handheld echolocation devices.
- 15 Alvin Lucier, *I am Sitting in a Room* (1993) [audio recording], USA: Lovely Music Ltd.
- 16 Lucier, *I am Sitting in a Room*.
- 17 Taken from William Carlos Williams, *Paterson*, ed. Christopher MacGowan (New York: New Directions Publishing, 1995). The line appears in several places in the book, e.g. in the poem 'The Delineaments of the Giants', 16.
- 18 For Lucier's own account of the genesis of this piece, on which I've drawn, cf. Alvin Lucier, 'My Affairs with Feedback', *Resonance* 9, no. 2 (2002): 24–5.
- 19 Lucier, 'My Affairs with Feedback', 24.
- 20 This is the title of the aforementioned short but impressive essay by her on the nature of music: see note 1.
- 21 Emmanuel Holterbach, 'Eliane Radigue: Feedback Works 1969–1970', in the liner notes for Eliane Radigue, *Feedback Works 1969–70* (2014) [audio recording], Italy: Alga Marghen, 3.
- 22 Eliane Radigue, 'Interview with Eliane Radigue' [video], interview by Evelyne Gayou, trans. David Vaughn, YouTube (recorded 2013, uploaded 12 November 2018), <https://www.youtube.com/watch?v=dByqwi7Jvbo&t=1088s> (accessed 6 February 2021, 10:44–11:00).
- 23 Radigue, 'Interview with Eliane Radigue', 2:36–3:03.
- 24 Holterbach, 'Eliane Radigue', 3.
- 25 For her description of her work habits at this time, cf. Eliane Radigue, 'Interview about the Feedback Works: Paris, 5 April 2011', interview by

Emmanuel Holterbach, in the liner notes for Eliane Radigue, *Feedback Works 1969–70* (2014) [audio recording], Italy: Alga Marghen, 8. Strictly speaking, the opening passages of *Opus 17*, her final work from this period, do not exclusively employ feedback in the usual sense. Rather, Radigue used techniques that were, unbeknownst to her, nearly identical to Lucier's in *I am Sitting in a Room*, sending excerpts of Chopin and Wagner through reinjection loops until they transformed into shimmering walls of ambience. However, in the remainder of *Opus 17*, as well as all of the works leading up to it, acoustic and tape feedback were Radigue's only 'instruments'. Cf. Emmanuel Holterbach, 'Eliane Radigue: Opus 17', in the liner notes to Eliane Radigue, *Opus 17* (2015) [audio recording], Italy: Alga Marghen.

- 26 Cf. 'Interview about the Feedback Works', as well as 'Eliane Radigue: Feedback Works 1969–70', 3–5.
- 27 Radigue, 'Interview about the Feedback Works'.
- 28 Radigue, 'Interview about the Feedback Works', 48.
- 29 Radigue, 'Interview about the Feedback Works', 47–8.
- 30 Radigue, 'Interview about the Feedback Works', 48.
- 31 Radigue, 'Interview about the Feedback Works', 48.
- 32 Radigue, 'Interview about the Feedback Works', 48.
- 33 Radigue, 'Interview about the Feedback Works', 49.